Sukruta Prakash **Midigeshi**

Applied Scientist · Amazon

Email | Website | Google Scholar | LinkedIn

Education

Indian Institute of Technology Gandhinagar — *Gandhinagar, Gujarat*
B.Tech. in Computer Science and Engineering — *Nov 2020 – Jun 2024*
- Cumulative Grade Point Average (CGPA): **9.07/10.0**
- **Dean's List** for academic excellence in semesters 2, 5, 6, 7 and 8
- Highest grade (**11/10**) in Natural Language Processing, in a class of ∼100 students

Experience

Amazon
Applied Scientist, Advisor: Dr. Saurabh — *Jun 2026 – Present*

Microsoft Research
Research Fellow, Advisor: Dr. Gaurav Sinha — *Jul 2024 – Jun 2026*

D. E. Shaw
Software Intern, Advisor: Mr. Devender Lega — *May 2023 – Jun 2023*

goGlocal
Software Intern, Advisor: Mr. Shubham Kedia — *Jun 2022 – Jul 2022*

eLitmus
Software Intern, Advisor: Mr. Shireesh Jayashetty — *Jan 2022 – Jun 2022*

Publications and Preprints

Packing Linear Programs and Fractional Knapsack using Comparison Oracles — *[arXiv]*
Ritabrata Barat$^{\alpha-\beta}$, Siddharth Barman$^{\alpha-\beta}$, Nirjhar Das$^{\alpha-\beta}$, **Sukruta Prakash Midigeshi**$^{\alpha-\beta}$ — *Jul 2026*
WINE 2026 Conference on Web and Internet Economics

Batched Contextual Slate Bandits with GLM Rewards — *[arXiv]*
Tanmay Goyal, **Sukruta Prakash Midigeshi**, Gaurav Sinha — *Jun 2026*
ICML 2026 International Conference on Machine Learning

Characterizing Deep Research: A Benchmark and Formal Definition — *[arXiv][Code]*
Abhinav Java*, Ashmit Khandelwal*, **Sukruta Prakash Midigeshi***, Aaron Halfaker, Amit Deshpande, Navin Goyal, Ankur Gupta, Nagarajan Natarajan, Amit Sharma — *Aug 2025*
ICLR 2026 International Conference on Learning Representations; also SEA Workshop @ NeurIPS 2025

Achieving Limited Adaptivity for Multinomial Logistic Bandits — *[RLJ][arXiv]*
Sukruta Prakash Midigeshi, Tanmay Goyal, Gaurav Sinha — *Aug 2025*
RLC 2025 Reinforcement Learning Conference

Plan-RAG: Planning-guided Retrieval Augmented Generation — *[arXiv]*
Prakhar Verma, **Sukruta Prakash Midigeshi**, Gaurav Sinha, Arno Solin, Nagarajan Natarajan, Amit Sharma — *Oct 2024*
Workshop on Reasoning and Planning for Large Language Models @ ICLR 2025

Improving Reachability on Reasoning Puzzles — *[OpenReview]*
Sukruta Prakash Midigeshi*, Sai Soumya Nalli*, Utkarsh Tiwari, Amit Deshpande, Nagarajan Natarajan, Vineeth N. Balasubramanian, Amit Sharma, Gaurav Sinha — *Apr 2026*
Workshop on Logical Reasoning of Large Language Models @ ICLR 2026

Note: $\alpha - \beta$ denotes alphabetical ordering, which is customary in theoretical computer science research; $*$ denotes equal contribution.

Research Projects

Achieving Limited Adaptivity for GLM Bandits with Large Action Spaces — *Microsoft Research*
With Dr. Gaurav Sinha and Tanmay Goyal — *Jan 2025 – Present*
- Designed soft-elimination algorithms for non-contextual GLM bandits with large action spaces; extending to the contextual case.

Approximating Neural Tangent Kernels using LSH — *IIT Gandhinagar*
With Prof. Anirban Dasgupta — *Aug 2023 – Nov 2023*
- Extended locality-sensitive-hashing techniques for kernel ridge regression beyond shift-invariant kernels to NTKs; derived theoretical bounds and compared approximation methods empirically.

Gossip: A Tree-walk and Byte-code Compiler in Python — *IIT Gandhinagar*
With Prof. Balagopal Komarath — *Jan 2022 – Apr 2022*
- Designed the syntax, grammar and parser; implemented a static type-checker, custom error handling, recursion support and a CLI.

Honors & Awards

2021–24 **Dean's List**, Academic excellence in semesters 2, 5, 6, 7 and 8 — *IIT Gandhinagar*
Highest Grade (11/10), Natural Language Processing, class of ∼100 students — *IIT Gandhinagar*

Teaching

Spring'24 **Teaching Assistant**, Machine Learning (Prof. Nipun Batra), ∼200 students — *IIT Gandhinagar*
Spring'24 **ADH Mentor**, Theory of Computing (Prof. Balagopal Komarath), ∼100 students — *IIT Gandhinagar*
Spring'23 **Teaching Assistant**, Discrete Mathematics (Prof. Bireswar Das), ∼60 students — *IIT Gandhinagar*

Relevant Coursework

Theory Discrete Mathematics [A], Theory of Computing [A-], Complexity Theory [A], Advanced Algorithms [A-], Combinatorics with Applications in CS [A]
Machine Learning Machine Learning [A], Deep Learning [A], Natural Language Processing [A+], Data Science [A]
Systems Compilers [A-]

Leadership & Service

Women in Tech (WiT) — *IIT Gandhinagar*
Founding Member
- Founded and led an initiative to increase female participation in computer science research and industry.

Inter-IIT Sports Meet — *IIT Gandhinagar*
Second Seed, Squash Team — *2022*
- Represented IIT Gandhinagar as the second seed on the squash team at the Inter-IIT Sports Meet.

Blithchron '21, Cultural Fest — *IIT Gandhinagar*
Design Coordinator — *2021*
- Led the design team and coordinated across all departments of the organizing committee.

Peer-Assisted Learning (PAL) — *IIT Gandhinagar*
PAL Mentor
- Ran regular mentorship sessions helping students improve their English language proficiency.